projection tracks the rise
Spearman ρ
0.5
0.0
−0.5
own direction
shift norm held fixed
1,000 random directions
0.8B
2B
4B
9B
27B
Llama
Gemma
Phi
inserted text, dose −1.5
refusal removed
0.4
0.2
0.0
own direction
random, norm-matched
random, KL-matched
0.8B
2B
4B
9B
27B
Llama
Gemma
Phi
original text, dose +1.5
refusal created
0.4
0.2
0.0
own direction
random, norm-matched
0.8B
2B
4B
9B
27B
Llama
Gemma
Phi